Noisy Labels in Classification

A Bayesian Framework applied to Occupational Coding

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Outline

Motivation

Noisy labels in occupational coding: Disagreement is information

"Provide counseling and support to families in crisis situations"

Two plausible codes for the same description

- **ISCO 2635** Social work professional
- **ISCO 3412** Social work associate

The challenge

- Multiple raters produce different codes
- No gold standard label is available

What the Bayesian D&S model does

- **Disagreement** modeled explicitly
- **Uncertainty** quantified

The Bayesian Dawid & Skene Model

Dawid & Skene model: Setup

- Each item i has **exactly one** unobserved true class $C_i \in \{1, \ldots, J\}$
- Each rater k has a confusion matrix $\Pi^{(k)}$
- Diagonal entries are per class accuracies

	Obs 1	Obs 2
True 1	$\boldsymbol{\pi_{11}^{(k)}}$	$\pi_{12}^{(k)}$
True 2	$\pi_{21}^{(k)}$	$\boldsymbol{\pi_{22}^{(k)}}$

Confusion matrix.

Dawid & Skene model: Likelihood of observed labels

- One true class per item
- 1) Class prevalence $p = (p_1, \ldots, p_J)$, with $\sum_{j=1}^{J} p_j = 1$
- 2) Rater confusion matrices $\Pi^{(k)} \in [0, 1]^{J \times J}$, with row sums equal to 1

Complete-data likelihood (latent T)

$$L_c(p, \pi; T, n)$$

Observed-data likelihood (marginalized over T)

$$L(p, \pi; n) \;=\; \sum_{T} L_c(p, \pi; T, n)$$

Frequentist version vs Bayesian extension

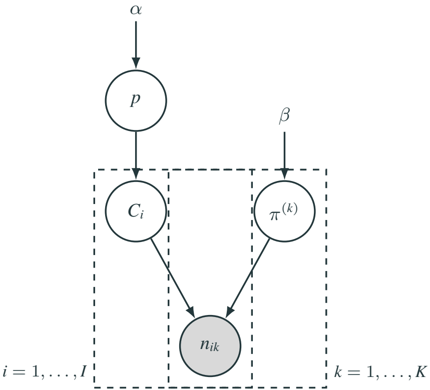
Frequentist Approach

- Principle: Treat T as missing data
- Iteratively maximize the likelihood with EM
- Output: point estimates $\hat{p}, \hat{\Pi}^{(k)}$
- Limited uncertainty quantification

Bayesian model

- Principle: combine likelihood with priors for p and $\Pi^{(k)}$
- Inference: given the data, sample via Monte Carlo
- Output: posterior distribution with Credible Intervals

Bayesian extension: Graphical model



- $p \sim \text{Dirichlet}(\alpha)$
- Each row of $\Pi^{(k)} \sim \text{Dirichlet}(\beta)$
- True class: $C_i \sim \text{Categorical}(p)$

Strong assumptions may be violated

Model assumptions

- **Uniqueness:** each item has one true class
- **Independence:** raters act independently given the latent true class
- **Stationarity:** rater error rates are constant across items

Practical hurdles in this setting

- **Ambiguity:** job texts are vague or mix tasks
- **House effects:** shared training or tools induce correlated errors
- **Process effects:** attention drifts, e.g., alphabetical sorting

Data and Disagreement

Data overview

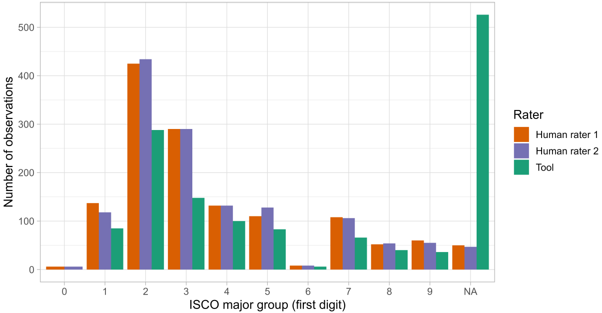
Dataset

- 1400 respondents
- 2 professional human coders
- 1 ML tool
- ISCO 08 and KldB 2010

Categories

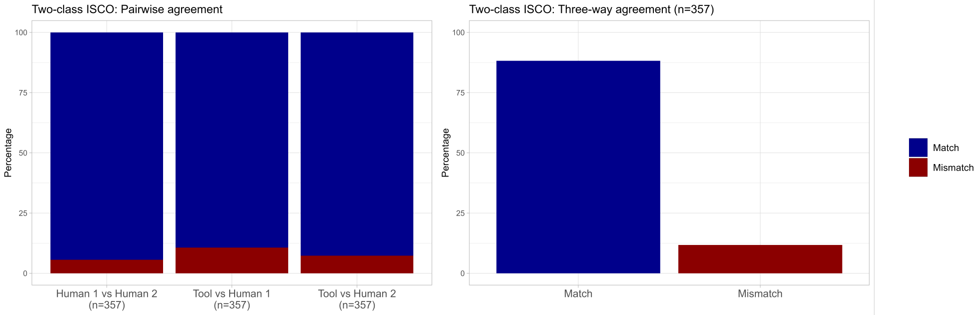
- ISCO 2: Professionals
- ISCO 3: Technicians and Associate Professionals

ISCO major groups



Full sample distribution

Observed agreement (ISCO 2 vs 3)

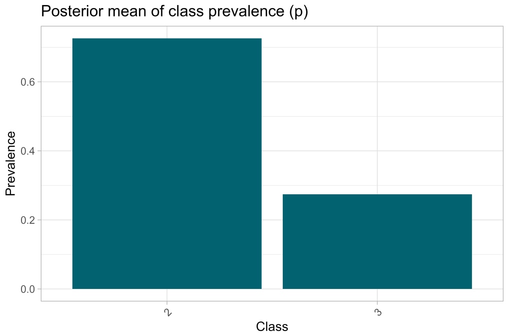


Subset: major groups 2 vs 3, $n = 357$

Results: ISCO, Two categories

The Baseline model: Uniform priors

- Baseline specification: $p \sim \text{Dirichlet}(1, 1)$, rows of $\Pi^{(k)} \sim \text{Dirichlet}(1, 1)$
- Output: posterior class prevalence for ISCO major groups 2 vs 3



Posterior mean: $p(\text{class } 2) \approx 0.726$, $p(\text{class } 3) \approx 0.274$

Rater confusion: Suggestions by the model

- Human rater 2 is strongest across both classes
- The tool shows an asymmetric weakness on the true 3 row, it overassigns class 2 when the true class is 3

Human rater 1

	obs 2	obs 3
true 2	0.953	0.047
true 3	0.068	0.932

Human rater 2

	obs 2	obs 3
true 2	0.987	0.013
true 3	0.021	0.979

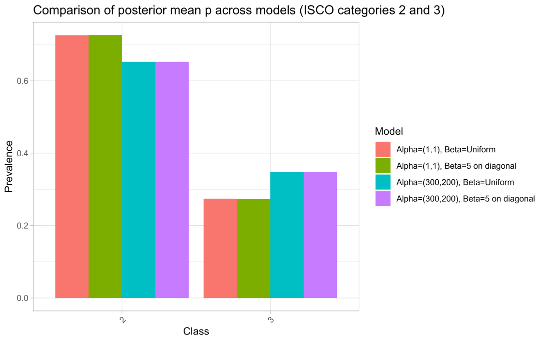
Tool

	obs 2	obs 3
true 2	0.953	0.047
true 3	**0.130**	0.870

Influence of Priors

Influence of priors: α has major impact

- **Prevalence prior:** $\alpha = (300, 200)$
  - Prior mean: 60% class 2 appearance
- **Confusion prior:** diagonal-favoring $\beta = 5$ (for $J = 2$)
  - Prior diagonal: 83% correct classification



But what about β ?

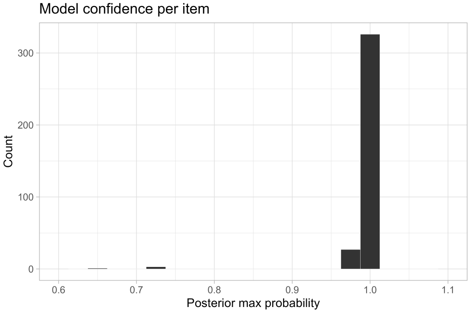
Uncertainty and triage

Item level uncertainty measures

Summary based on $\omega_{ij} = P(C_i = j \mid \text{data})$.

- Maximum posterior probability:

$$m_i = \max_j \omega_{ij}$$

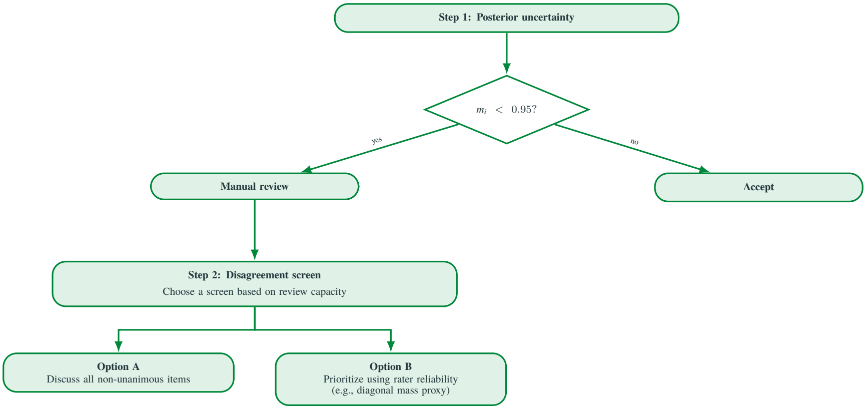


Most uncertain ISCO items (baseline model)

		Posterior statistics		Ratings		
Item ID	$\hat{C}_i$	m_i	H_i	**H1**	**H2**	Tool
285	3	0.650	0.647	**3**	**2**	3
132	2	0.734	0.579	**2**	**3**	2
234	2	0.734	0.579	**2**	**3**	2
307	2	0.734	0.579	**2**	**3**	2

$\rightarrow$ High uncertainty coincides with human disagreement (**H1** vs **H2**) only.

Triage workflow for manual review



Conclusion

Takeaways

- Capability: D&S Model makes rater disagreement explicit.
- Interpretation: Strong assumptions underlie the the present setting
- Bayesian version: Outputs uncertainty at item and rater level
- Practical triage: flag a small fraction of high-uncertainty cases for manual review.
- Next: add covariates and difficulty, validate triage in practice and extend to finer ISCO levels.

Scaling from two to three categories

From two classes to three classes

Parameter growth

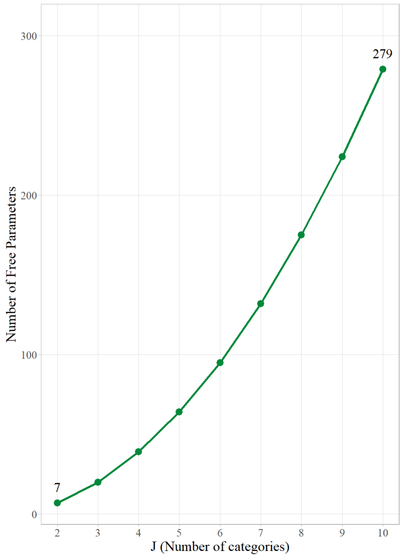
Count of free parameters:

$$(J-1)(1+KJ)$$

- $J=2, K=3:$ 7 parameters
- $J=3, K=3:$ 20 parameters

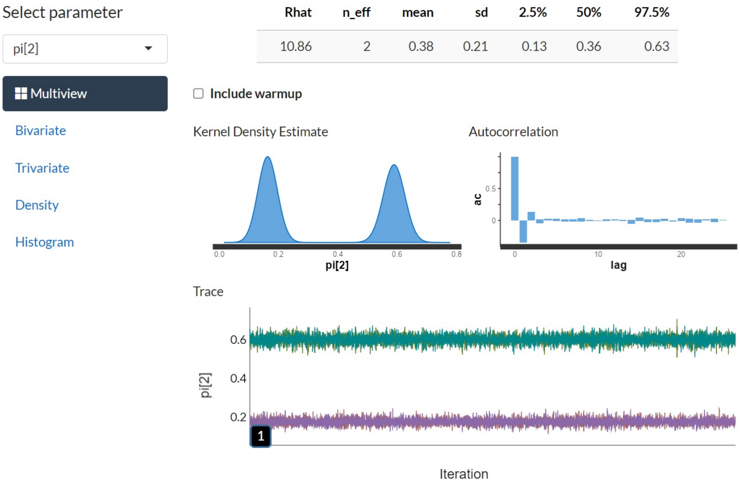
Label switching

- Likelihood is symmetric under label permutations
- Chains can explore different labelings



Free parameters vs J (for $K=3$)

Label switching in practice



Split-$\widehat{R}$ and low ESS indicate non-identification due to label switching.

Fix: scale diagonal prior mass with J

Assume a row-wise Dirichlet prior with diagonal mass β_{diag} and off-diagonal mass β_{off}. Then

$$q := \mathbb{E}\left[\pi_{jj}^{(k)}\right] \;=\; \frac{\beta_{\text{diag}}}{\beta_{\text{diag}} + (J-1)\beta_{\text{off}}}.$$

Fix a target diagonal expectation q, solve for β_{diag}:

$$\beta_{\text{diag}} \;=\; \frac{q}{1-q}(J-1)\beta_{\text{off}}.$$

Example with $\beta_{\text{off}} = 1$ and $q = 0.83$:

- $J = 2$: $\beta_{\text{diag}} \approx 5$
- $J = 3$: $\beta_{\text{diag}} \approx 10$

After stronger regularization (diagonal $\beta = 15$)

Select parameter

pi[2]

Rhat	n_eff	mean	sd	2.5%	50%	97.5%
1	16000	0.59	0.02	0.55	0.59	0.64

Multiview

Bivariate

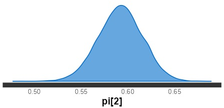
Trivariate

Density

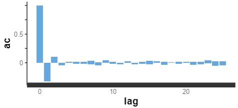
Histogram

Include warmup

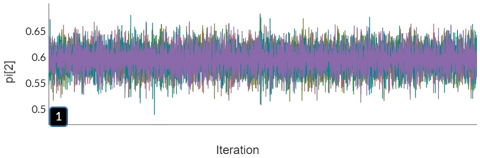
Kernel Density Estimate



Autocorrelation



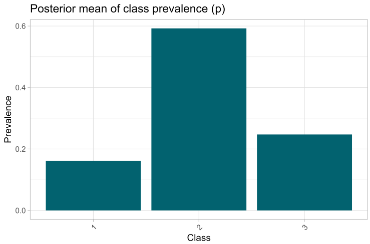
Trace

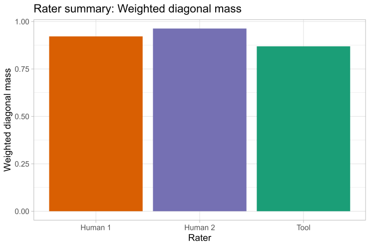


Use your mouse to highlight areas in the traceplot to zoom into. Double-click to reset. The number in the small black box in the bottom left corner controls the *roll period*. If you specify a roll period of N the resulting graph will be a moving average, with each plotted point representing the average of N points in the data.

Diagnostics stabilize once the diagonal prior mass is increased.

Results: ISCO three category extension





Same qualitative pattern: human rater 2 remains strongest, the tool struggles more on minority classes